Write a program to calculate the total number of context switches needed if the operating system implements a shortest job first (preemptive) scheduling algorithm.

```
  GNU nano 7.2                                    sjf.py
# Simple SJF (Preemptive) Scheduling

arrival = [0, 2, 6]       # arrival times
burst   = [10, 20, 30]    # burst times
n = len(arrival)

remaining = burst[:]      # copy burst times
time = 0
completed = 0
last = -1
context_switches = 0
completion = [0] * n

while completed < n:
    # find process with shortest remaining among arrived
    idx = -1
    shortest = 999999
    for i in range(n):
        if arrival[i] <= time and remaining[i] > 0:
            if remaining[i] < shortest:
                shortest = remaining[i]
                idx = i

    if idx == -1:
        time += 1
        continue
```

```
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    if idx == -1:
        time += 1
        continue

    if last != -1 and last != idx:
        context_switches += 1
    last = idx

    remaining[idx] -= 1
    time += 1

    if remaining[idx] == 0:
        completion[idx] = time
        completed += 1

waiting = []
for i in range(n):
    turnaround = completion[i] - arrival[i]
    waiting.append(turnaround - burst[i])

avg_waiting = sum(waiting) / n

print("Total Context Switches:", context_switches)
print("Average Waiting Time:", avg_waiting)
```

OUTPUT:-

```
students@M309-37: ~
students@M309-37:~$ nano sjf.py
students@M309-37:~$ python3 sjf.py
  File "/home/students/sjf.py", line 21
    idx = i
IndentationError: unexpected indent
students@M309-37:~$ python3 sjf.py
  File "/home/students/sjf.py", line 21
    idx = i
IndentationError: unexpected indent
students@M309-37:~$ nano sjf.py
students@M309-37:~$ python3 sjf.py
Total Context Switches: 2
Average Waiting Time: 10.666666666666666
students@M309-37:~$
```